



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Data Visualisation with Tableau

TGS Ref No: TGS-2020503177

Conducted by

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Version 17.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
17.0	15 August 2026	Two-day lesson plan aligned to the current visual deck, ten Tableau labs, 60-minute WA and 60-minute PP.	Dr Alfred Ang

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Course Overview

Data Visualisation with Tableau is a two-day, 16-hour WSQ programme aligned to Data Visualization (ICT-DIT-3006-1.1). Learners progress through a connected Meridian Retail SG scenario using Tableau Desktop or Tableau Public and realistic synthetic Excel sources.

Course Code	TGS-2020503177
Training Duration	16 hours across 2 days
Assessment	WA 60 minutes; PP 60 minutes
Trainer	Dr Alfred Ang
Mode	Instructor-led classroom with individual Tableau labs

Learning Outcomes

- LO1: Apply basic Tableau features for displaying information.
- LO2: Synthesise various plots to present data visually.
- LO3: Transform data to create informative and dynamic data displays.
- LO4: Create dashboards and scorecards to display internal and external benchmark data.
- LO5: Apply calculations and parameters to create interactive graphics, visuals and technical features.
- LO6: Perform analytics and communicate data limitations and interpretations of findings.

Daily Schedule

Day 1: Foundations, data modelling, chart choice and transformation

Time	Duration	Topic / Activity	Method	Slides
09:00-09:20	20 min	Welcome, AM digital attendance, introductions and outcomes	Facilitated briefing	—
09:20-10:30	70 min	Topic 1: Tableau ecosystem, interface, field roles and hierarchies	Concept briefing, examples and questioning	Slides 18-36
10:30-10:40	10 min	Tea break	Break	—
10:40-12:30	110 min	Labs 1–2: source connection, field roles and drill paths	Demonstration, individual practice and debrief	Slides 32-35
12:30-13:30	60 min	Lunch break	Lunch	—
13:30-14:30	60 min	Topic 2: connections, extracts, data modelling and chart choice	Concept briefing, examples and questioning	Slides 37-55
14:30-15:20	50 min	Labs 3–4: relationships, extracts and decision charts	Demonstration, individual practice and debrief	Slides 51-54
15:20-15:30	10 min	Tea break	Break	—
15:30-16:25	55 min	Topic 3: Data Interpreter, reshaping, filter order, groups and sets	Concept briefing, examples and questioning	Slides 56-74
16:25-17:40	75 min	Lab 5: clean, split and pivot campaign data	Demonstration, individual practice and debrief	Slides 70-71
17:40-18:00	20 min	Day 1 reflection, Q&A and PM digital attendance	Reflection and Q&A	—

Scheduled time excluding lunch: 480 minutes (8 hours).

Day 2: Dashboards, interactivity, calculations, analytics and assessment

Time	Duration	Topic / Activity	Method	Slides
09:00-09:15	15 min	Day 1 recap and AM digital attendance	Facilitated briefing	—
09:15-10:20	65 min	Topic 4: dashboard hierarchy, scorecards, actions and stories	Concept briefing, examples and questioning	Slides 76-94
10:20-10:30	10 min	Tea break	Break	—
10:30-12:30	120 min	Labs 6–8: cohorts, executive dashboard, actions and story	Demonstration, individual practice and debrief	Slides 72-93
12:30-13:30	60 min	Lunch break	Lunch	—
13:30-14:15	45 min	Topic 5: calculations, table calculations, parameters and dynamic views	Concept briefing, examples and questioning	Slides 95-111
14:15-14:55	40 min	Lab 9: dynamic profitability analysis	Demonstration, individual practice and debrief	Slides 109-110
14:55-15:05	10 min	Tea break	Break	—
15:05-15:25	20 min	Topic 6: reference context, trends, forecasts, clusters and limitations	Concept briefing, examples and questioning	Slides 112-128
15:25-15:50	25 min	Lab 10: analytics and responsible communication	Demonstration, individual practice and debrief	Slides 126-127
15:50-16:00	10 min	Assessment briefing and digital attendance	Individual summative assessment	—
16:00-17:00	60 min	Written Assessment (WA/SAQ), open book	Individual summative assessment	—
17:00-18:00	60 min	Practical Performance (PP), open book	Individual summative assessment	—

Scheduled time excluding lunch: 480 minutes (8 hours).

Topic-by-Topic Delivery Guide

Topic 01: Basic Tableau Features • Slides 18-36

Tableau ecosystem · interface · data roles · shelves · hierarchy

- **Visual analysis cycle:** Connect, explore, explain and refine around a decision question.
- **Dimensions and measures:** Dimensions slice the view; measures are aggregated to quantify it.
- **Discrete and continuous:** Blue headers create categories; green axes create a continuous range.
- **Workbook structure:** Worksheets build views; dashboards combine them; stories sequence insights.
- **Fields and metadata:** Names, types, roles, aliases and default properties shape every view.
- **Hierarchy:** Ordered levels let users drill from overview to operational detail.

Lab 1: Connect and Profile Service Operations Data • Slides 32-33

Alignment: LO1 · K1 · A1. Learner output: A connected Tableau workbook with validated field roles and a service-volume view by region and channel. Acceptance: The workbook connects without broken fields; dates and numbers have correct roles; the view reconciles to the Excel record count.

Lab 2: Build Hierarchies and Drill Paths • Slides 34-35

Alignment: LO1 · K1 · A1. Learner output: A geographic and product hierarchy with a drillable sales-and-margin view. Acceptance: Both hierarchies drill in the correct order and totals remain stable at every displayed level.

Topic 02: Data Sources and Visualisation • Slides 37-55

Connections · extracts · relationships · joins · blending · chart choice

- **Connection strategy:** Choose live or extract by freshness, performance and governance needs.
- **Relationships:** Preserve each table's grain and combine data at analysis time.
- **Joins:** Physically combine rows; cardinality and unmatched keys determine the result.
- **Visual encoding:** Position and length usually support more accurate comparisons than area or colour.
- **Chart purpose:** Use bars for comparison, lines for time, scatterplots for relationship and treemaps for hierarchy.
- **Aggregation:** SUM, AVG, COUNT and distinct count answer different business questions.

Lab 3: Relate Orders, Products and Returns • Slides 51-52

Alignment: LO2 · K3 · A2. Learner output: A governed multi-table data model plus an extract and reconciliation sheet.

Acceptance: Order totals are not duplicated, unmatched keys are visible, and the extract refreshes from the supplied workbook.

Lab 4: Choose and Build Decision Charts • Slides 53-54

Alignment: LO2 · K3 · A2. Learner output: Four fit-for-purpose worksheets: ranked bar, quarterly line, profit-sales scatterplot and category treemap. Acceptance: Every chart answers its stated question, uses truthful axes, and includes enough context to interpret the marks.

Topic 03: Data Transformation • Slides 56-74

Data Interpreter · split · merge · pivot · filters · groups · sets

- **Analysis-ready structure:** One row is one observation; one column is one variable; one table is one grain.
- **Data Interpreter:** Removes presentation clutter from spreadsheets while preserving the underlying evidence.
- **Split and pivot:** Separate compound fields and reshape wide columns into analysis-ready rows.
- **Filter order:** Extract, source, context, dimension and measure filters do not behave identically.
- **Groups:** Combine members into a stable business category.
- **Sets:** Define a dynamic in/out population that can drive comparisons and actions.

Lab 5: Clean, Split and Pivot Campaign Data • Slides 70-71

Alignment: LO3 · K4 · A3. Learner output: An analysis-ready long table with clean location fields and documented transformation checks. Acceptance: The transformed data has one row per campaign-month, clean location fields, no blank campaign key and reconciled spend totals.

Lab 6: Filter, Group and Set Customer Segments • Slides 72-73

Alignment: LO3 · K4 · A3. Learner output: A segment view using context filters, grouped categories and a top-customer set. Acceptance: The high-value set responds to the selected period and the business grouping is explainable from the source fields.

Topic 04: Dashboards and Stories • Slides 76-94

KPI scorecards · layout · actions · device design · story sequence

- **Four Ws:** Who is the audience, what decision, when is it used and where is action taken?
- **Dashboard hierarchy:** Lead with decision KPIs, then trends, drivers and diagnostic detail.
- **Layout containers:** Tiled containers create predictable alignment; floating objects need deliberate control.
- **Actions:** Filter, highlight, URL, navigation, parameter and set actions create purposeful interaction.
- **Benchmark context:** Targets, prior periods and peer ranges turn values into performance signals.
- **Stories:** A sequence of story points communicates change, evidence and decision.

Lab 7: Build an Executive Performance Dashboard • Slides 90-91

Alignment: LO4 · K5 · A4. Learner output: A 16:9 scorecard dashboard with KPI tiles, trend, ranked outlets, benchmark context and visible filter state. Acceptance: The dashboard is readable at presentation size, totals reconcile, active filters are visible and every element supports a named decision.

Lab 8: Add Actions and Tell a Performance Story • Slides 92-93

Alignment: LO4 · K5 · A4. Learner output: A dashboard with filter/highlight/navigation actions and a five-point performance story. Acceptance: Actions work from the intended source sheets, preserve context and the story ends with an evidence-backed action.

Topic 05: Calculations and Parameters • Slides 95-111

Calculated fields · table calculations · logic · parameters · dynamic views

- **Row-level calculation:** Evaluated for each record before aggregation.
- **Aggregate calculation:** Combines measures at the view's level of detail.
- **Table calculation:** Computed over the marks already returned to the visualisation.
- **Logical design:** IF, IIF and CASE translate business rules into transparent classifications.
- **Parameters:** Single values controlled by the user; pair them with calculations or actions to change behaviour.
- **Dynamic view:** A parameter can switch metrics, thresholds, reference values or dimensions.

Lab 9: Create Dynamic Profitability Analysis • Slides 109-110

Alignment: LO5 · K2 · A5. Learner output: A dynamic metric view with calculated fields, parameter controls, table calculations and a threshold filter. Acceptance: Metric and threshold controls change the view correctly, formulas reconcile to Excel controls and invalid denominator cases are handled.

Topic 06: Analytics and Communication • Slides 112-128

Reference lines · trends · forecasts · clusters · limitations · data stories

- **Reference context:** Average, median, target and distribution bands help the audience judge performance.
- **Trend models:** A fitted line summarises association; it does not establish causality.
- **Forecast:** Tableau extends a time series under model assumptions and expresses uncertainty.
- **Clustering:** Groups similar marks from selected features; business labels require interpretation.
- **Limitations:** Data coverage, quality, granularity, bias and model assumptions bound the conclusion.
- **Responsible narrative:** Separate observation, interpretation, recommendation and uncertainty.

Lab 10: Analyse Trends, Forecasts and Clusters • Slides 126-127

Alignment: LO6 · K1 · A6. Learner output: An analytics workbook with reference context, trend, forecast, clustering and a limitations-led decision narrative. Acceptance: The workbook states assumptions and limitations, distinguishes association from causation, and reports at least one action with a monitoring trigger.

Resources Required

- Tableau Desktop or Tableau Public on each learner laptop.
- Learner Guide and ten numbered lab folders.
- Starter Excel workbook plus `.twb` and `.twbx` sample for every lab.
- Projector for concept briefings and demonstrations.
- LMS/TMS access for courseware, attendance and assessment submission.

Assessment

- Written Assessment (WA/SAQ) — 5 open-ended questions, 60 minutes, open book.
- Practical Performance (PP) — 6 connected Tableau tasks, 60 minutes, open book.
- Assessment runs from 4:00 PM to 6:00 PM on Day 2.
- Assessment is individual and open book. Learners submit the completed papers and required Tableau evidence through the LMS.
- The new papers mirror the current v11 instrument type, question/task count, K/A mapping and timing.